

Q1.

An insurance company groups its vehicle insurance policies into two categories, car insurance and motorbike insurance.

The number of claims in a random sample of 80 policies was monitored and the results summarised in contingency **Table 1**.

Table 1

		Number of claims				
		0	1	2	3 or more	Total
Type of insurance policy	Car	9	10	11	5	35
	Motorbike	19	13	8	5	45
	Total	28	23	19	10	80

The insurance company decides to carry out a χ^2 -test for association between number of claims and type of insurance policy using the information given in **Table 1**.

- (a) The contingency table shown in **Table 2** gives some of the exact expected frequencies for this test.

Complete **Table 2** with the missing exact expected values.

Table 2

		Number of claims			
		0	1	2	3 or more
Type of insurance policy	Car		10.0625		4.375
	Motorbike			10.6875	

©2025 Exam Papers Practice. All rights reserved.

- (b) Carry out the insurance company's test, using the 10% level of significance.

(2)
(8)
(Total 10 marks)

Q2.

A dairy industry researcher, Robyn, decided to investigate the milk yield, classified as low, medium or high, obtained from four different breeds of cow, A, B, C and D.

The milk yield of a sample of 105 cows was monitored and the results are summarised in contingency **Table 1**.

		Yield			
		Low	Medium	High	Total
Breed	A	4	5	12	21
	B	10	6	4	20

	C	8	17	7	32
	D	5	20	7	32
	Total	27	48	30	105

The sample of cows may be regarded as random.

Robyn decides to carry out a χ^2 -test for association between milk yield and breed using the information given in **Table 1**.

- (a) Contingency **Table 2** gives some of the expected frequencies for this test.

Complete **Table 2** with the missing expected values.

Table 2		Yield		
		Low	Medium	High
Breed	A			6
	B	5.14	9.14	5.71
	C			
	D	8.23	14.63	9.14

(2)

- (b) (i) For Robyn's test, the test statistic $\sum \frac{(O - E)^2}{E} = 19.4$ correct to three significant figures.

Use this information to carry out Robyn's test, using the 1% level of significance.

(5)

- (ii) By considering the observed frequencies given in **Table 1** with the expected frequencies in **Table 2**, interpret, in context, the association, if any, between milk yield and breed.

(2)

(Total 9 marks)

Q3.

Students at a science department of a university are offered the opportunity to study an optional language module, either German or Mandarin, during their second year of study.

From a sample of 50 students who opted to study a language module, 31 were female. Of those who opted to study Mandarin, 8 were female and 12 were male.

Test, using the 5% level of significance, whether choice of language is independent of gender.

The sample of students may be regarded as random.

(Total 8 marks)

Q4.

It is claimed that the way in which students voted at a particular general election was independent of their gender.

In order to investigate this claim, 480 male and 540 female students who voted at this general election were surveyed. These students may be regarded as a random sample.

The **percentages** of males and females who voted for the different parties are recorded in the table.

	Conservative	Labour	Liberal Democrat	Other parties
Male	32.5	30	25	12.5
Female	40	25	20	15

- (a) Complete the contingency table below.

(2)

- (b) Hence determine, at the 1% level of significance, whether the way in which students voted at this general election was independent of their gender.

	Conservative	Labour	Liberal Democrat	Other parties	Total
Male					480
Female					540
Total					1020

(9)

©2025 Exam Papers Practice. All rights reserved.

(Total 11 marks)

Q5.

Year 12 students at Newstatus School choose to participate in one of four sports during the Spring term.

The students' choices are summarised in the table.

	Squash	Badminton	Archery	Hockey	Total
Male	5	16	30	19	70
Female	4	20	33	53	110
Total	9	36	63	72	180

- (a) Use a χ^2 test, at the 5% level of significance, to determine whether the choice of sport is independent of gender.

(10)

- (b) Interpret your result in part (a) as it relates to students choosing hockey.

(2)

(Total 12 marks)

Q6.

The time taken to complete a tennis match varies. Boris is a professional tennis player who believes that, because of his excellent physical condition, the longer a match lasts the better his chance of winning. An analysis of a sample of matches in which Boris had taken part showed that:

of 43 matches which lasted less than two hours, Boris had won 28;
of 52 matches which lasted two hours or longer, Boris had won 22.

- (a) Form the data into a 2 x 2 contingency table suitable for analysis using a χ^2 distribution.

(3)

- (b) Analyse the table that you have formed in part (a) using the 5% significance level.

(9)

- (c) Interpret your results in parts (a) and (b) in the context of the question.

(2)

(Total 14 marks)

Q7.

The board of an insurance company was concerned about the poor performance of its motor insurance business. Belinda was asked to collect and analyse data on the claims record of the 4902 customers who held motor policies in 2003.

- (a) The following contingency table summarises the number of claims made in 2003 classified by length of time the policy had been held.

		Number of Claims in 2003			
		0	1	2	3 or more
Policy held for	less than 2 years	1478	212	21	6
	2 or more years	2830	334	20	1

Use a χ^2 distribution and the 1% significance level to investigate whether the number of claims made in 2003 is associated with the length of time for which the policy had been held.

(11)

- (b) Belinda also constructed the following tables concerning the 4902 customers.

Table A

% of Total Number of Policy Holders who in 2003

		Made a Claim	Did not make a Claim
Gender	Male	8.4	57.6
	Female	3.7	30.3

Table B

		Total Number of Policy Holders in 2003	Number of Policy Holders who made a claim in 2003
Age (years)	< 25	956	280
	25 or more	3946	314

Table C

		Number of Policy Holders who in 2003	
		Did not make a Claim	Made a claim for more than £2000
Age (years)	< 25	676	145
	25 or more	3632	192

For **each** table:

- (i) state why, in its present form, it cannot be analysed as a contingency table using a χ^2 distribution; (3)
- (ii) construct a new table which can be analysed as a contingency table using a χ^2 distribution. **Do not analyse any of the tables you construct.**

(6)

(Total 20 marks)